Leveraging Eutectic Interfaces in Geodynamic models of Europa's Ice shell to constrain Physical Parameters

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Introduction

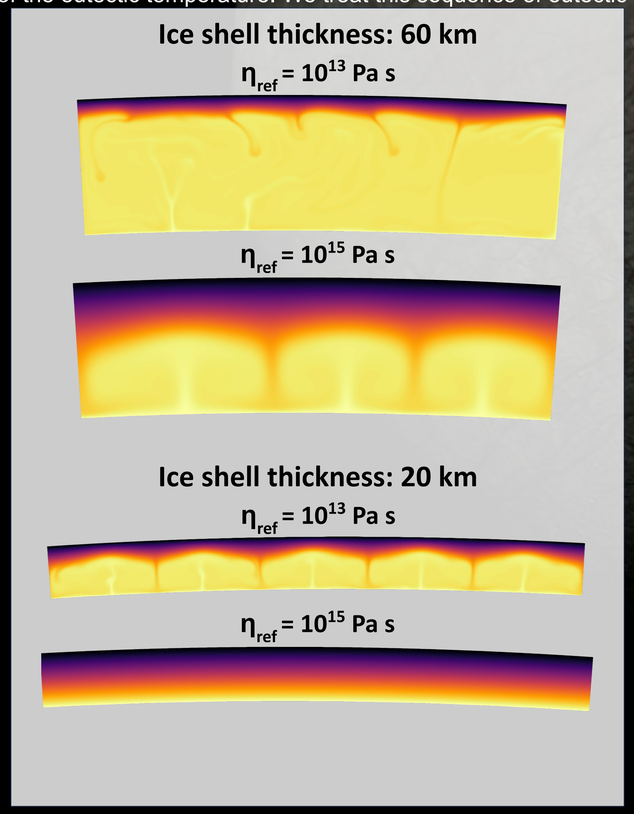
- Jupiter's moon Europa is one of the prime targets for planetary exploration due to its high astrobiological potential.
- The thickness of Europa's ice shell is poorly constrained with values of <1 km to up to 90 km [1-2].
- Ice-penetrating radars on NASA's Europa Clipper (REASON, [3]) and ESA's JUICE (RIME, [4]) missions aim to determine the thickness of Europa's ice shell.
- Recent studies have suggested that constraints on the thickness of Europa's ice shell can be obtained through the detection of eutectic interfaces, defined as the depth where brine becomes thermodynamically stable in the ice shell [5].

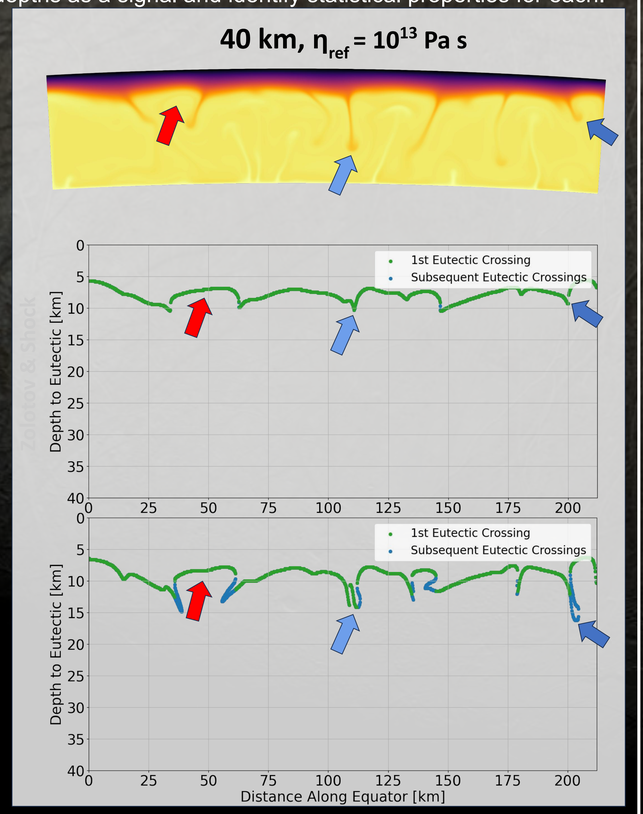
GAIA Simulations & Isotherm Identification

The depth of the eutectic interfaces depends on the thermal state of the ice shell, which is closely linked to the ice shell viscosity and large-scale dynamics [6]. We used the geodynamic code GAIA [7] to investigate the ice shell dynamics on Europa and performed a total of **40** simulations varying:

- The ice shell thickness (20 - 60 km)
- Ice shell viscosity that largely affect the convection pattern (10^{12} - 10^{15} Pa s)
- Eutectic temperature (binary end-members NaCl: 252 K and $MgSO_4$: 269 K, and modeled Europan ocean: 237K)
- The inclusion of **grain boundary sliding**, b**asal slip**, and **dislocation** as rheological deformation mechanisms
- **Thermal conductivity** (**K**) and **expansivity** (**α**)

Once a simulation has reached a statistical (quasi-)steady state, we determine the eutectic pattern by identifying the depths of the eutectic temperature. We treat this sequence of eutectic depths as a signal and identify statistical properties for each.

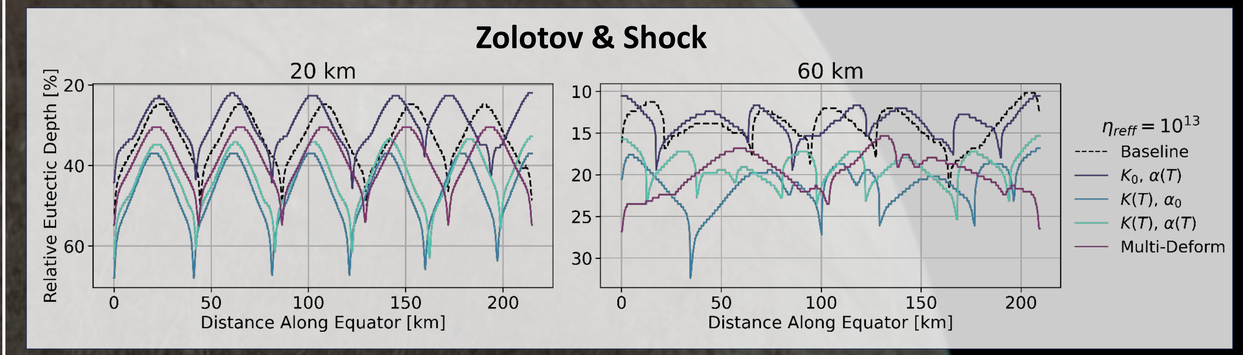




Eutectic Isotherm Analysis

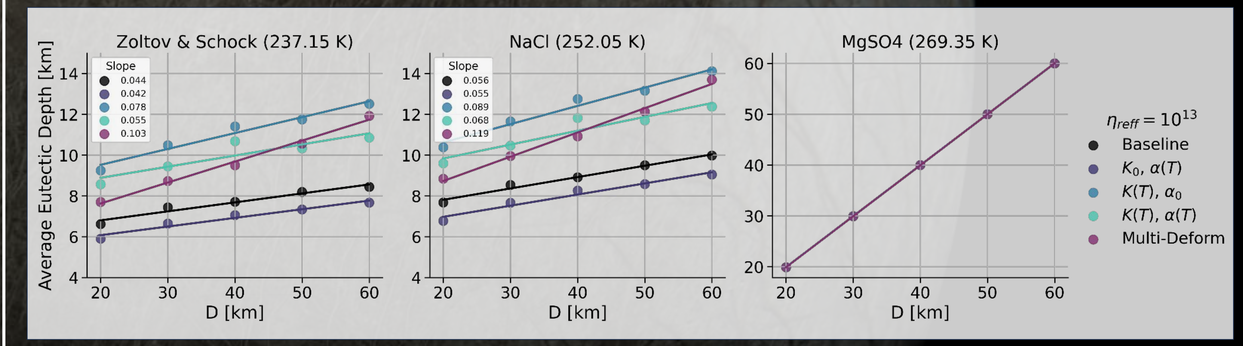
- All initial instances of eutectic crossing are treated as a "signal", estimating perfect radar melt detection
- Collect representative statistical data for each "signal"
- Includes the average eutectic depth, number of upwellings/downwellings (peaks/troughs)

*Only the Zolotov & Shock eutectics and those simulations at 1e13 reference viscosity are plotted below



Constraining Ice Shell Thickness

- Strong correlation between the eutectic depth and ice shell thickness for all viscosity values
- The depth to the eutectic interfaces increases with increasing ice shell thickness and decreases with decreasing viscosity. This trend is consistent across multiple eutectics
- The $MgSO_4$ eutectic temperature (269 K) is frequently reached at the ice-ocean boundary for all simulations



Conclusion

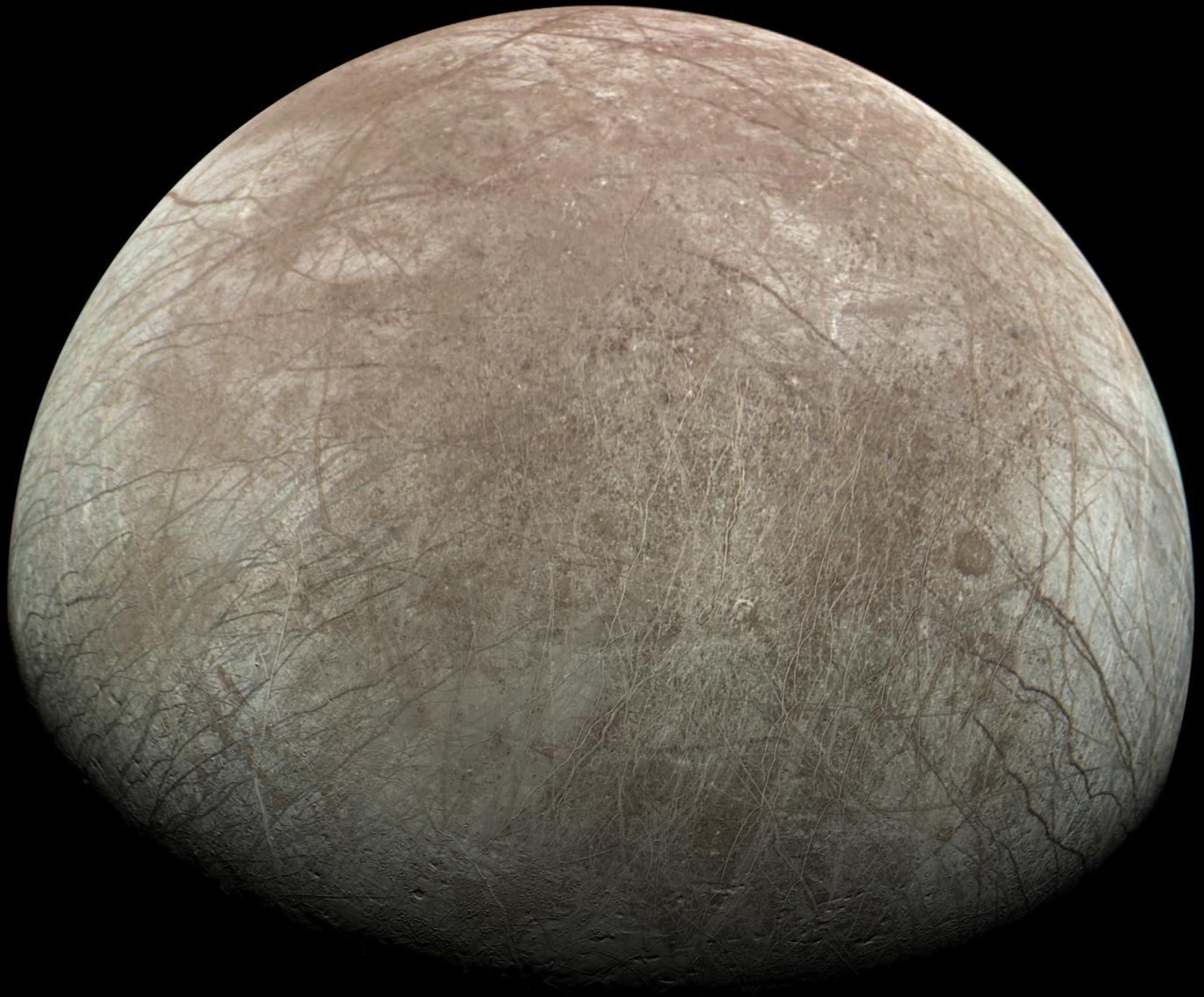
- Initial parameter study suggests ice shell thickness + shell viscosity constraints are possible
- A scaling law could be derived to relate **average eutectic depth** and **ice shell thickness**
- Trend exists across all identified eutectic isotherms that lie above the ice-ocean boundary

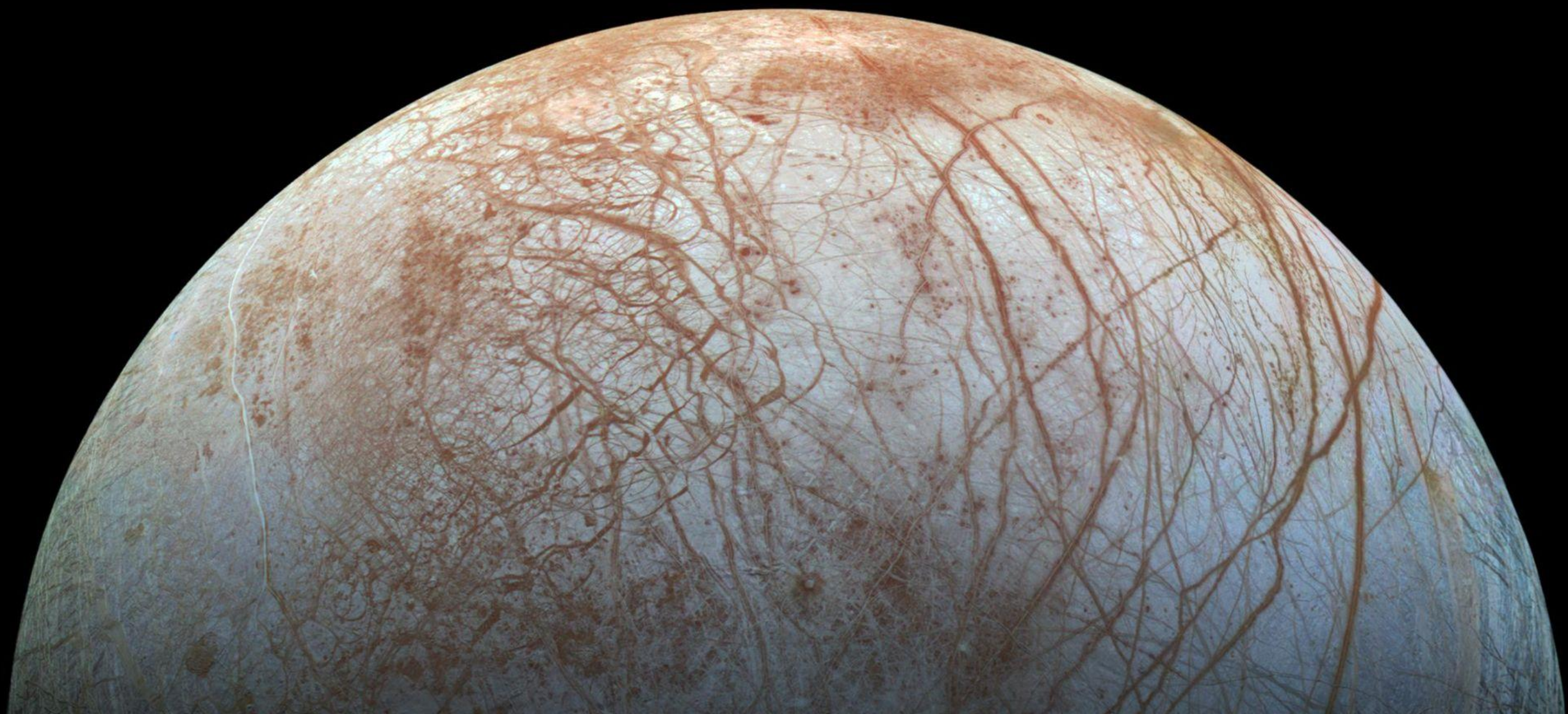
Future Work

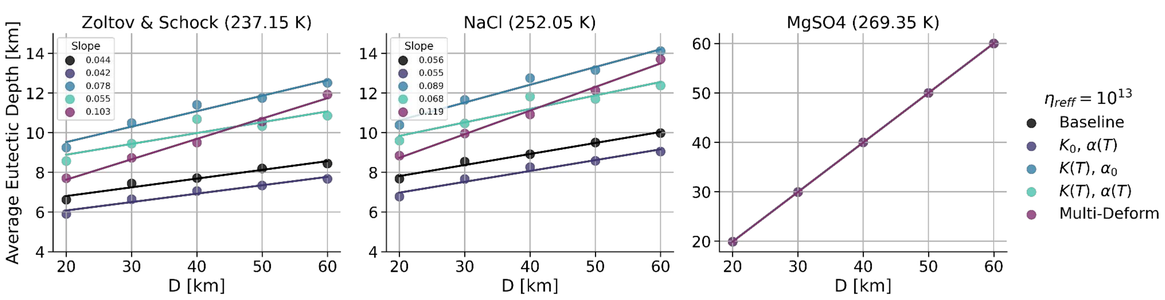
- Verify consistency of the results with higher resolution of simulations
- Add interpolation to better estimate the depth to eutectic temperature within GAIA
- Attempt to identify similar signal characteristics in a more "sparse" domain

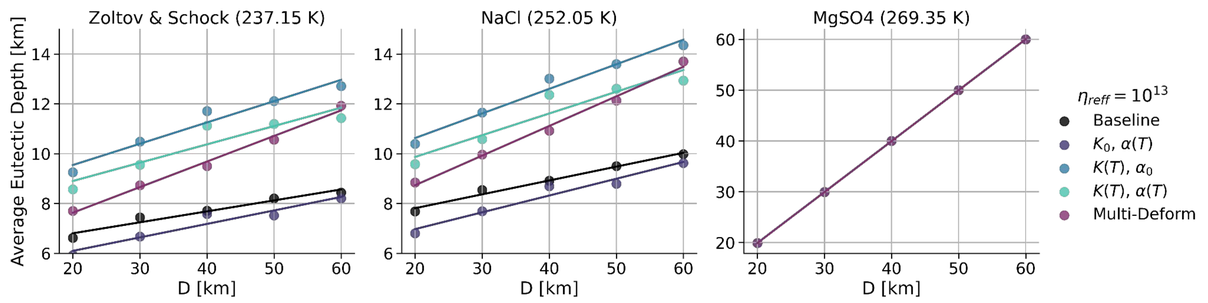
References: [1] Billings and Kattenhorn et al., 2005. [2] Vilella et al., 2020. [3] Blankenship et al., 2024. [4] Bruzzone et al., 2013. [5] Schroeder et al., 2024. [6] Kalousova et al., 2017. [7] Hüttig et al., 2013. [8] Wolfenbarger et al., 2022. [9] Zolotov & Shock et al., 2001.

20 Km






Zoltov & Schock (237.15 K)
NaCl (252.05 K)
MgSO4 (269.35 K)
Average Eutectic Depth [km]
D [km]
Slope
0.044
0.042
0.078
0.055
0.103
0.056
0.055
0.089
0.068
0.119
$\eta_{reff} = 10^{13}$
Baseline
K_0, $\alpha(T)$
$K(T)$, α_0
$K(T)$, $\alpha(T)$
Multi-Deform


Zoltov & Schock (237.15 K)
NaCl (252.05 K)
MgSO4 (269.35 K)
Average Eutectic Depth [km]
D [km]
$\eta_{reff} = 10^{13}$
Baseline
K_0, $\alpha(T)$
$K(T)$, α_0
$K(T)$, $\alpha(T)$
Multi-Deform